

Edmond Karp Algorithm

Module 4

Edmond Karp Algorithm

- **Edmonds–Karp algorithm** is an optimized implementation of the Ford–Fulkerson method for computing the maximum flow in a flow network
- The algorithm is identical to the Ford–Fulkerson algorithm, except that the search order when finding the augmenting path is defined
- The path found must be a shortest path from **S** to **T** that has available capacity and can be found by a **breadth-first search**, where we apply a weight of 1 to each edge
- One property of this algorithm is that the length of the shortest augmenting path increases monotonically.

Time Complexity of Ford Fulkerson Vs Edmonds Karp

The Ford–Fulkerson method using a DFS to find augmenting paths takes $O(Ef)$ where E is the number of edges and f is the max flow.

The Edmonds–Karp algorithm uses a **Breadth First Search (BFS)** to find augmenting paths which yields an arguably better time complexity of $O(VE^2)$. The major difference in this approach is that the time complexity no longer depends on the capacity value of any edge!

EDMONDS-KARP ALGORITHM(G,s,t)

begin

 initialise flow f to 0

while there exists a shortest augmenting path p in
 the residual network G_f **do**

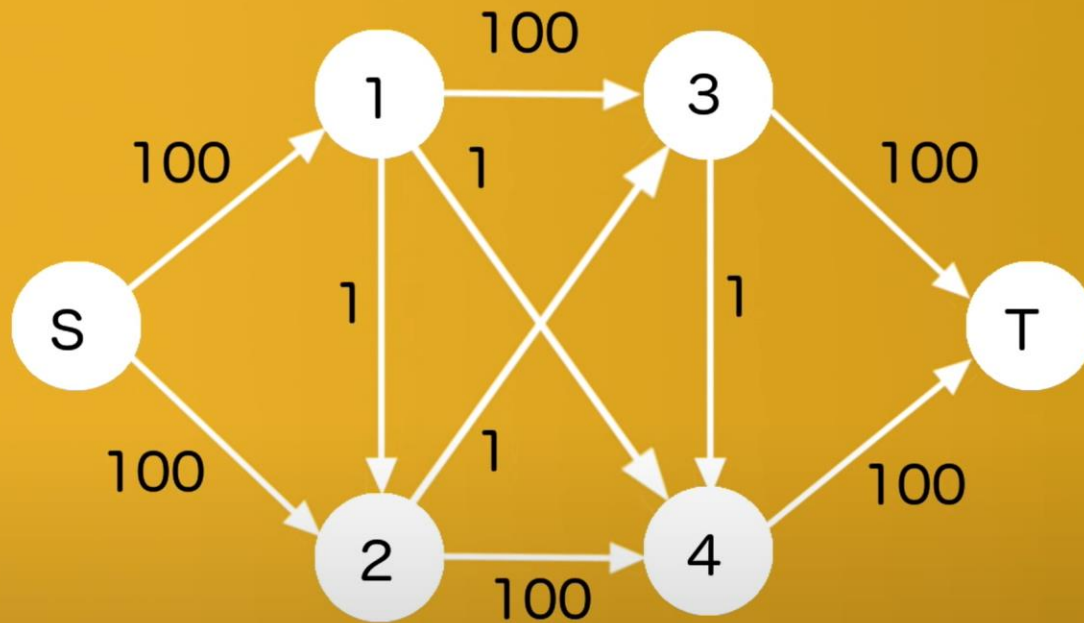
 augment flow f along p

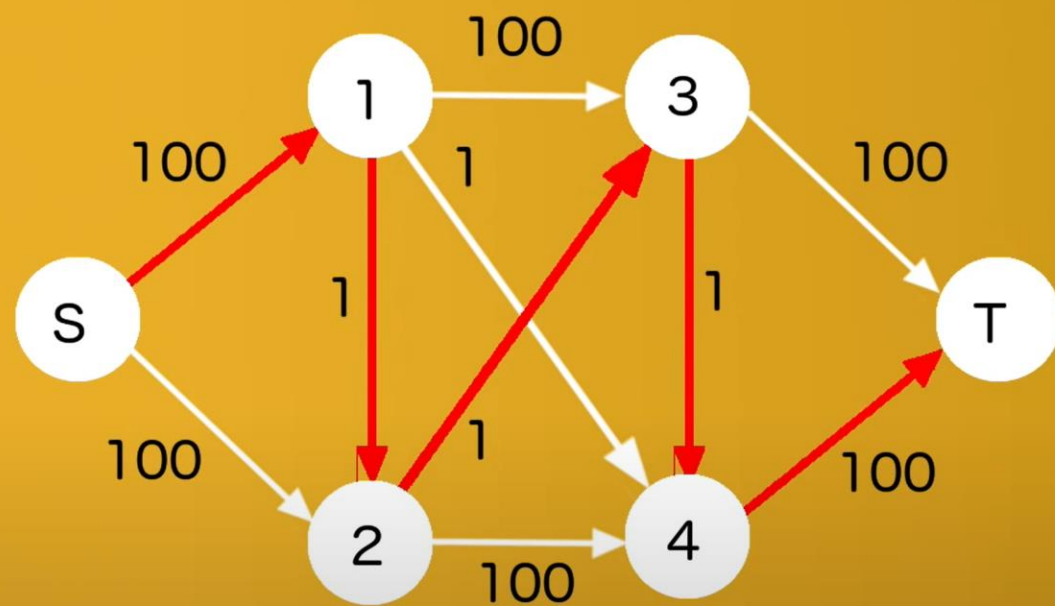
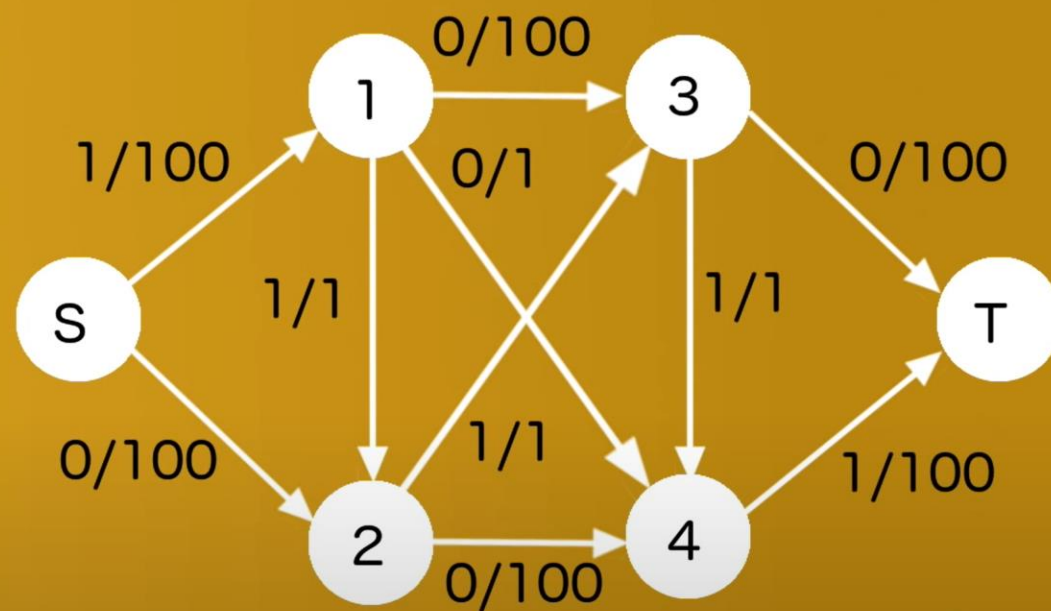
end

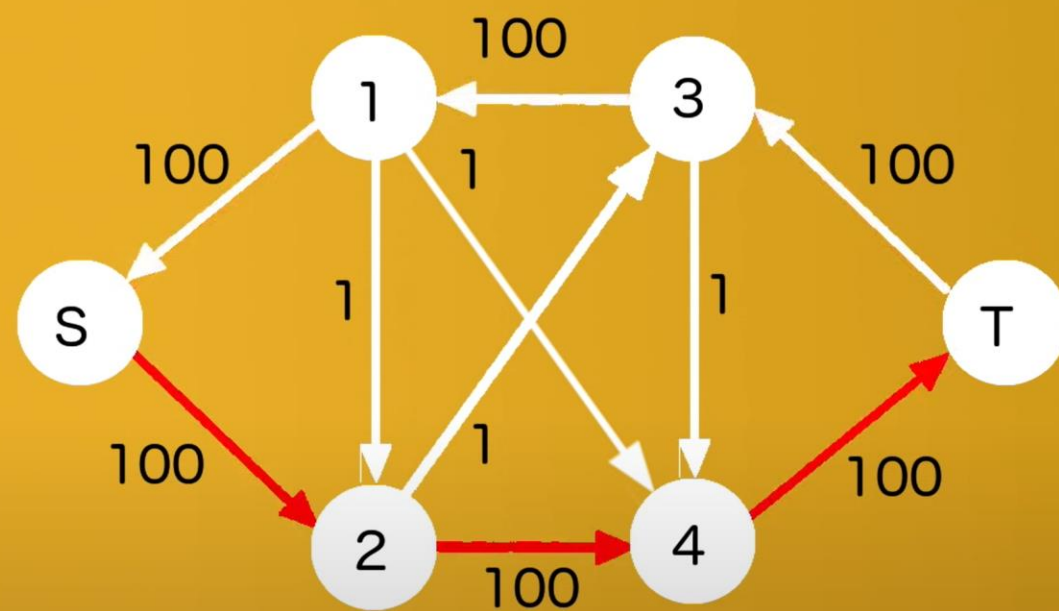
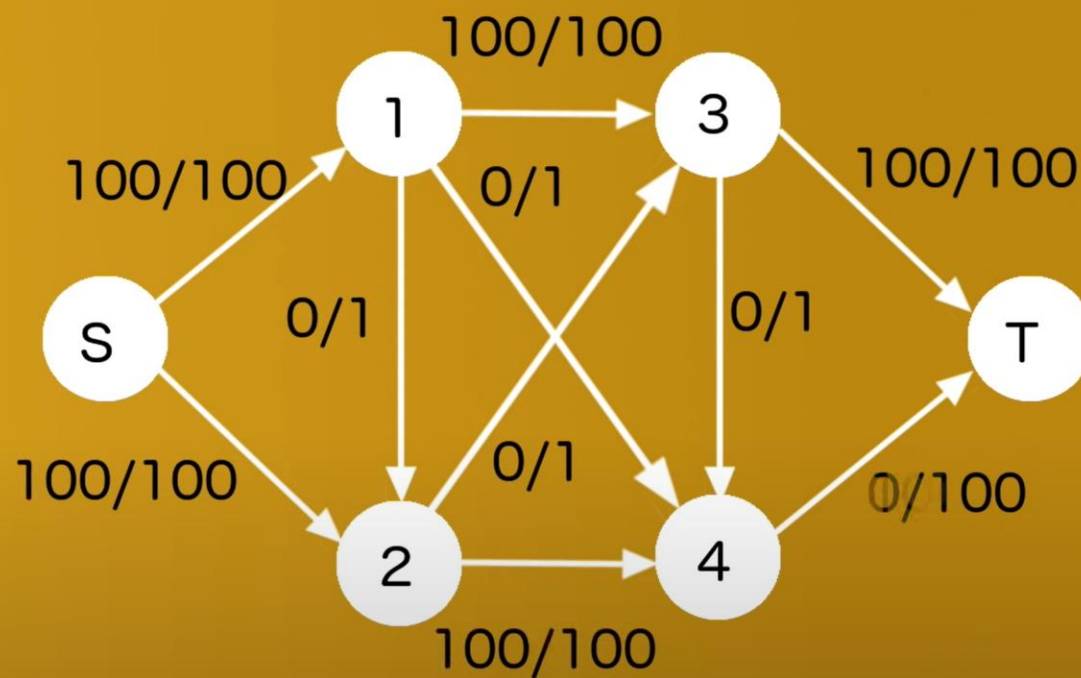
end

Need of Edmond Karp Algorithm

G



G_f  f 

G_f  f 

- Every iteration of the while loop, the shortest path algorithm must be performed on the residual network. This takes $O(E)$ time using breadth first search.

EDMONDS-KARP ALGORITHM(G,s,t)

begin

 initialise flow f to 0

while there exists a shortest augmenting path p in
 the residual network G_f **do**

 augment flow f along p

end

end

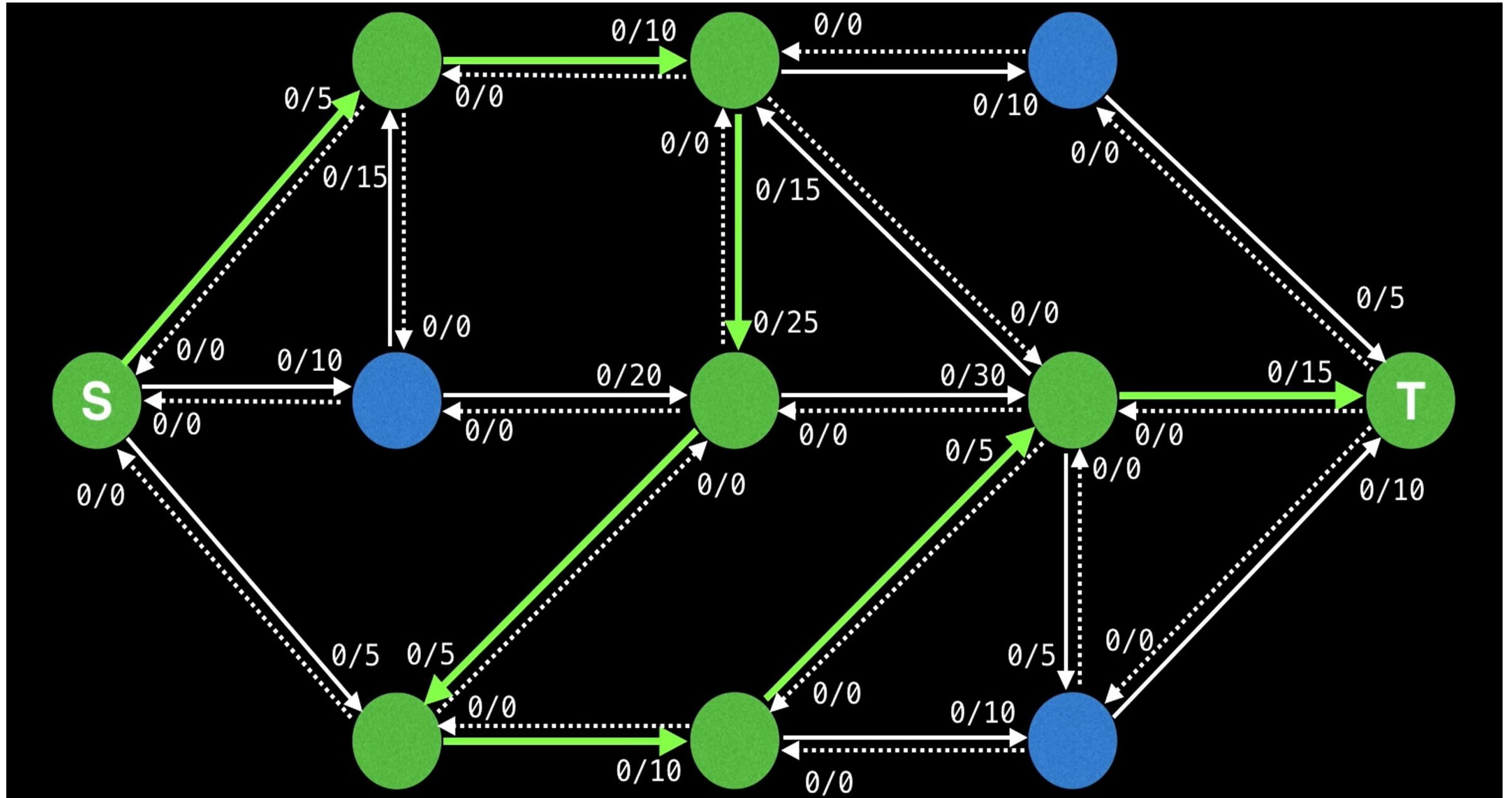


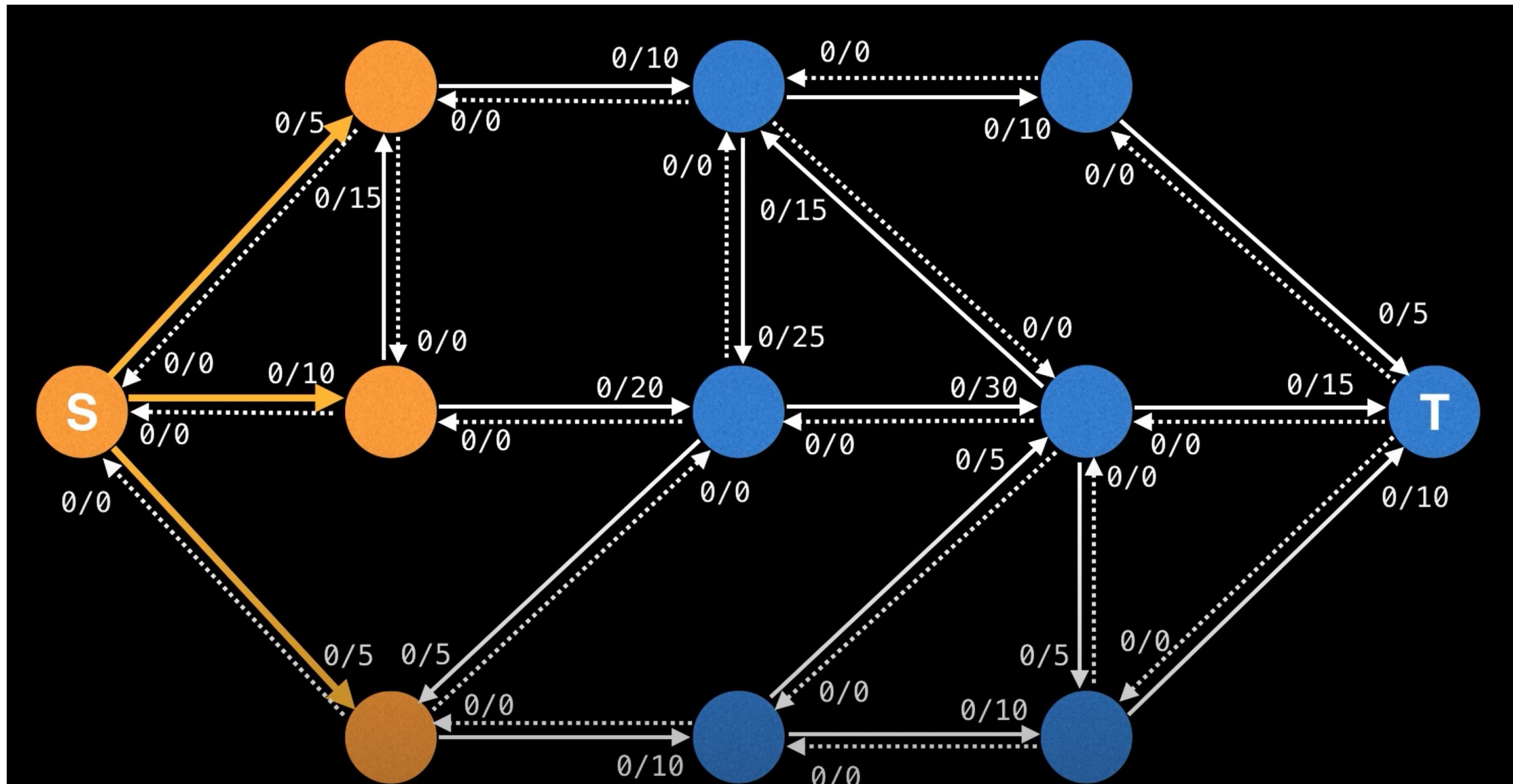
- What is the maximum amount of times the while loop can be executed?
- At every iteration one of the edges must become critical.
- From the time it is critical to the time it is critical again the distance to it must increase.
- It can increase at most V times as this is the longest path possible.
- This can happen to all E edges. Therefore, in the worst case E edges can become critical V times. So the loop can run VE times max.

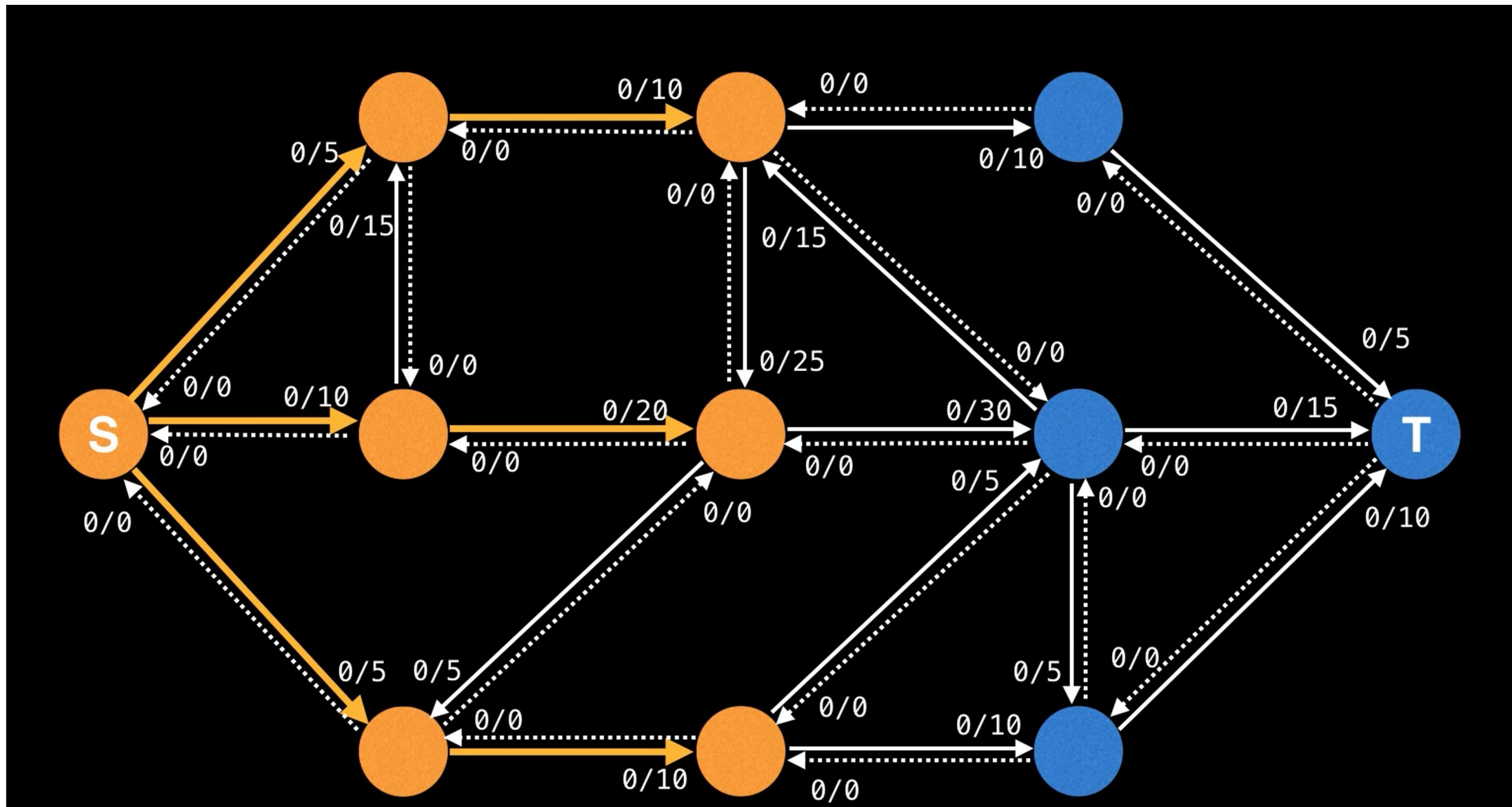
Therefore, running a loop that takes $O(E)$ time at most $O(VE)$ times means the algorithm has a runtime of

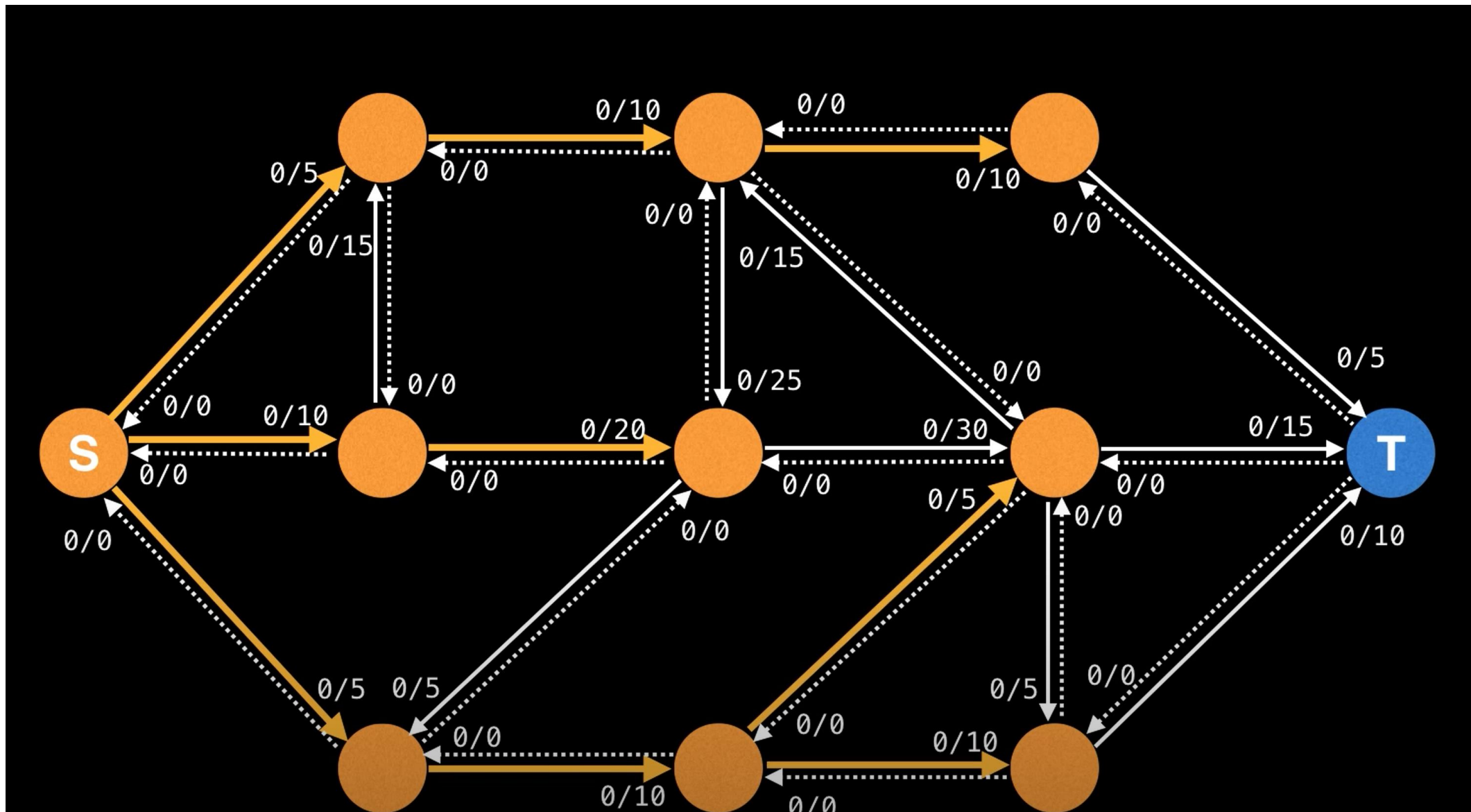
$$O(VE^2)$$

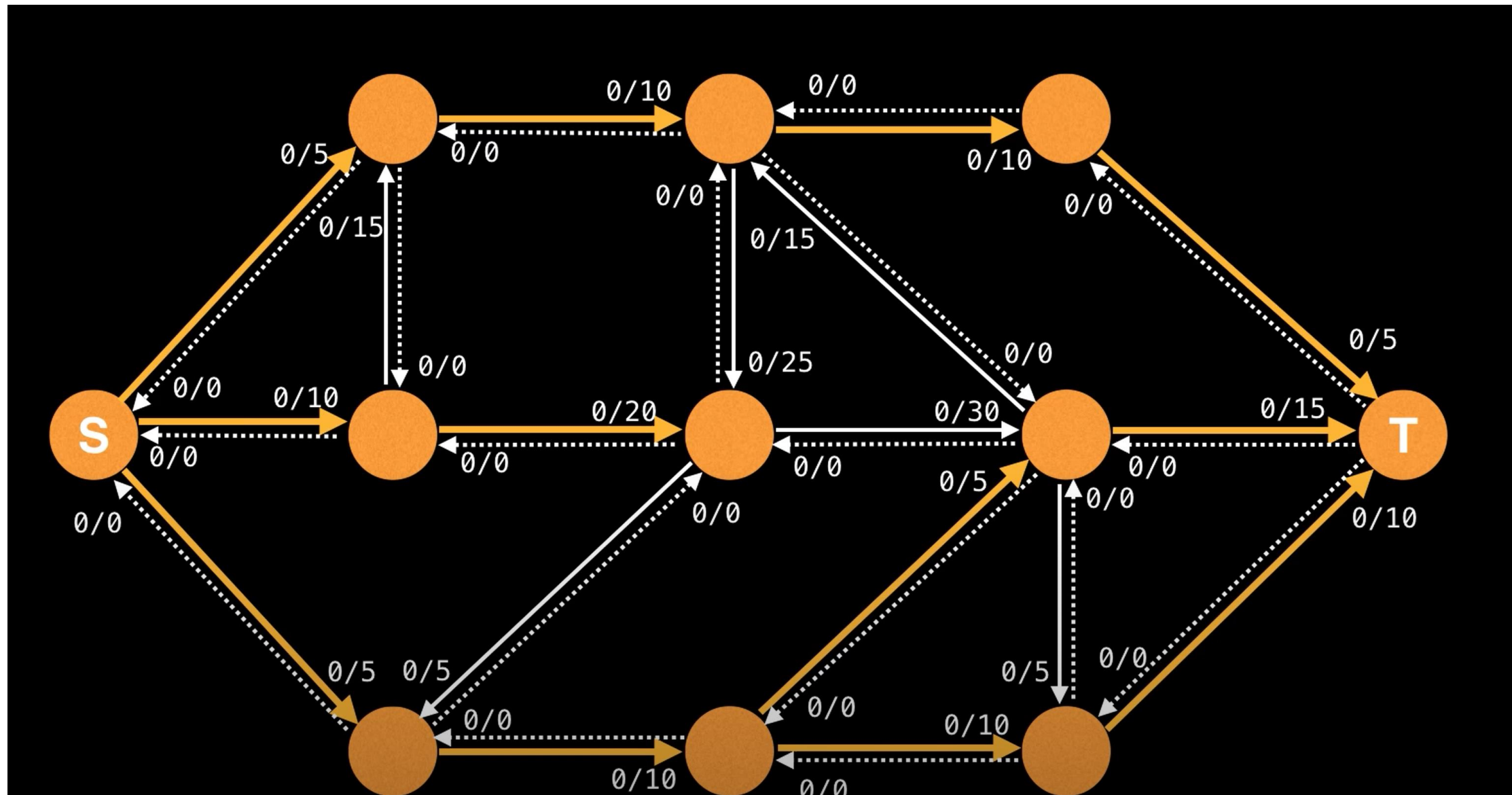
Augmenting path in Ford Fulkerson method

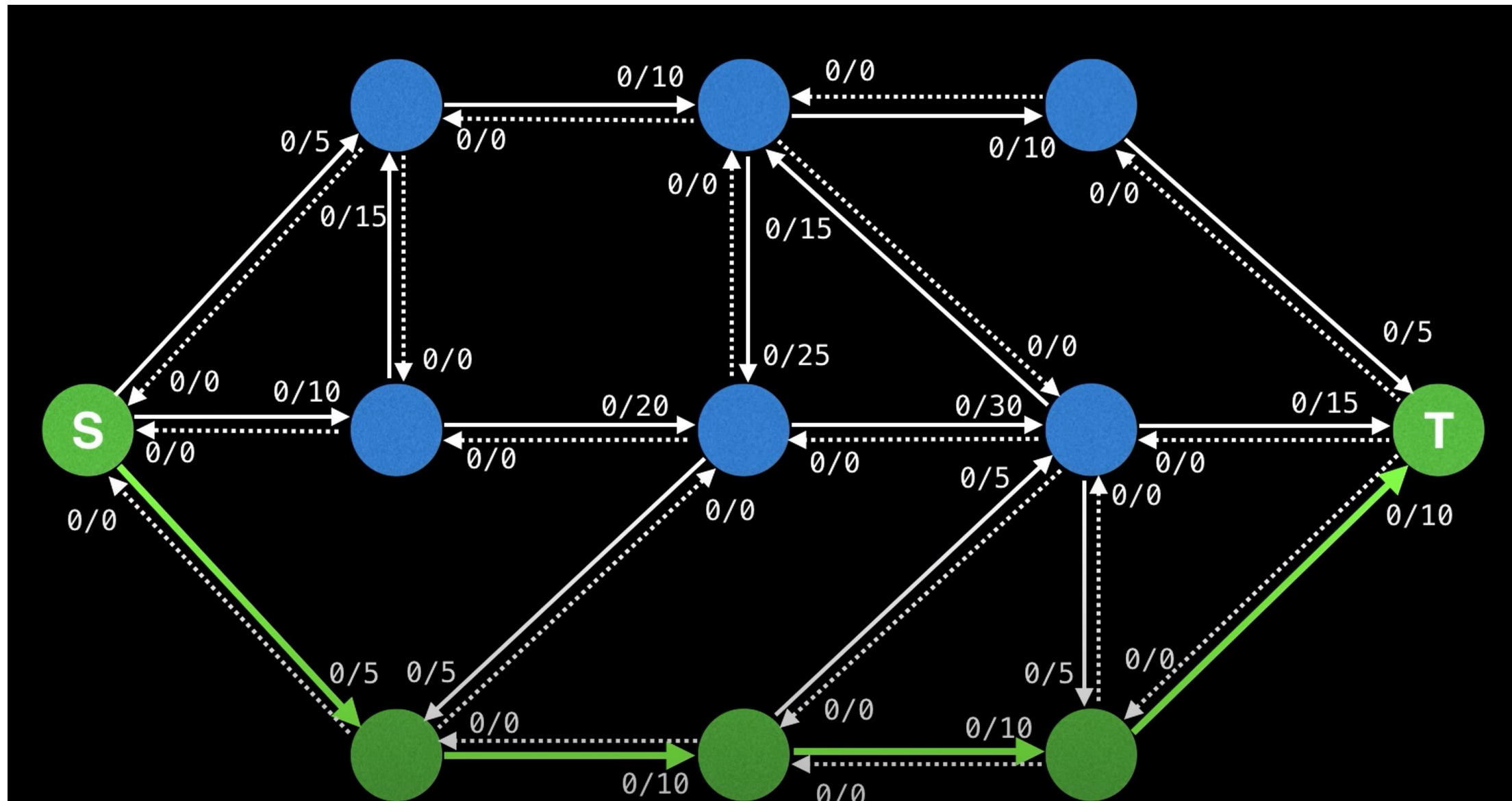


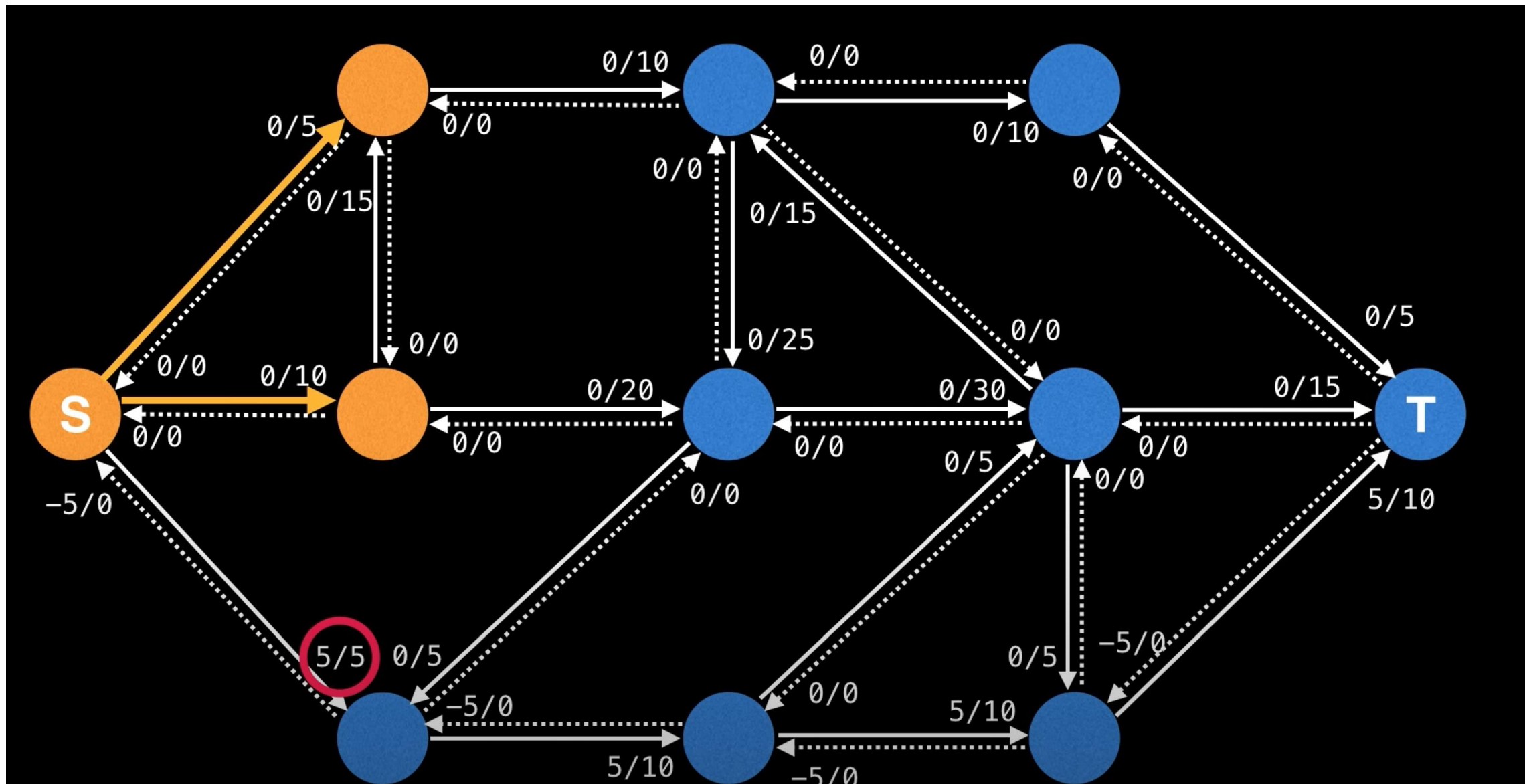


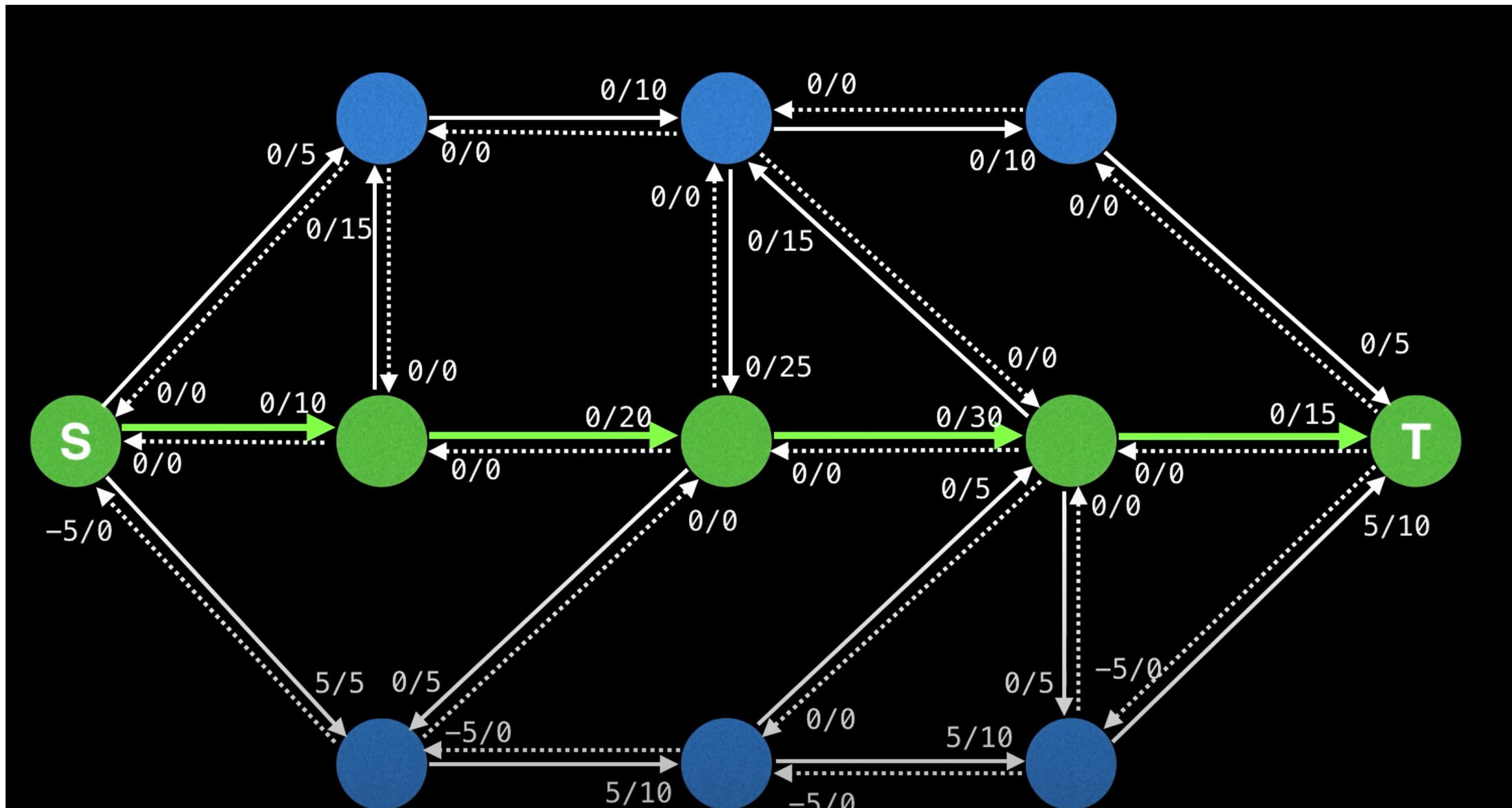


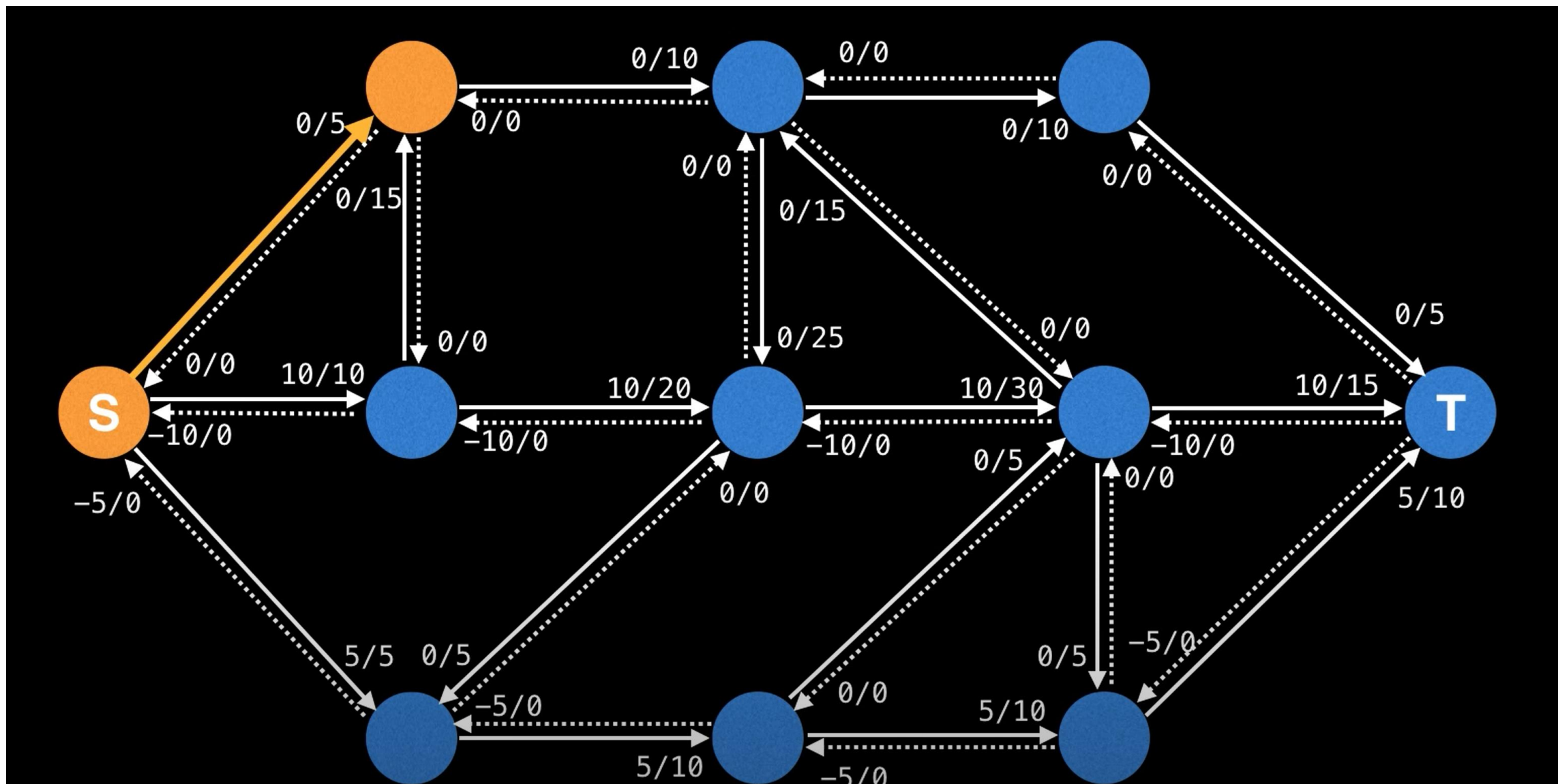


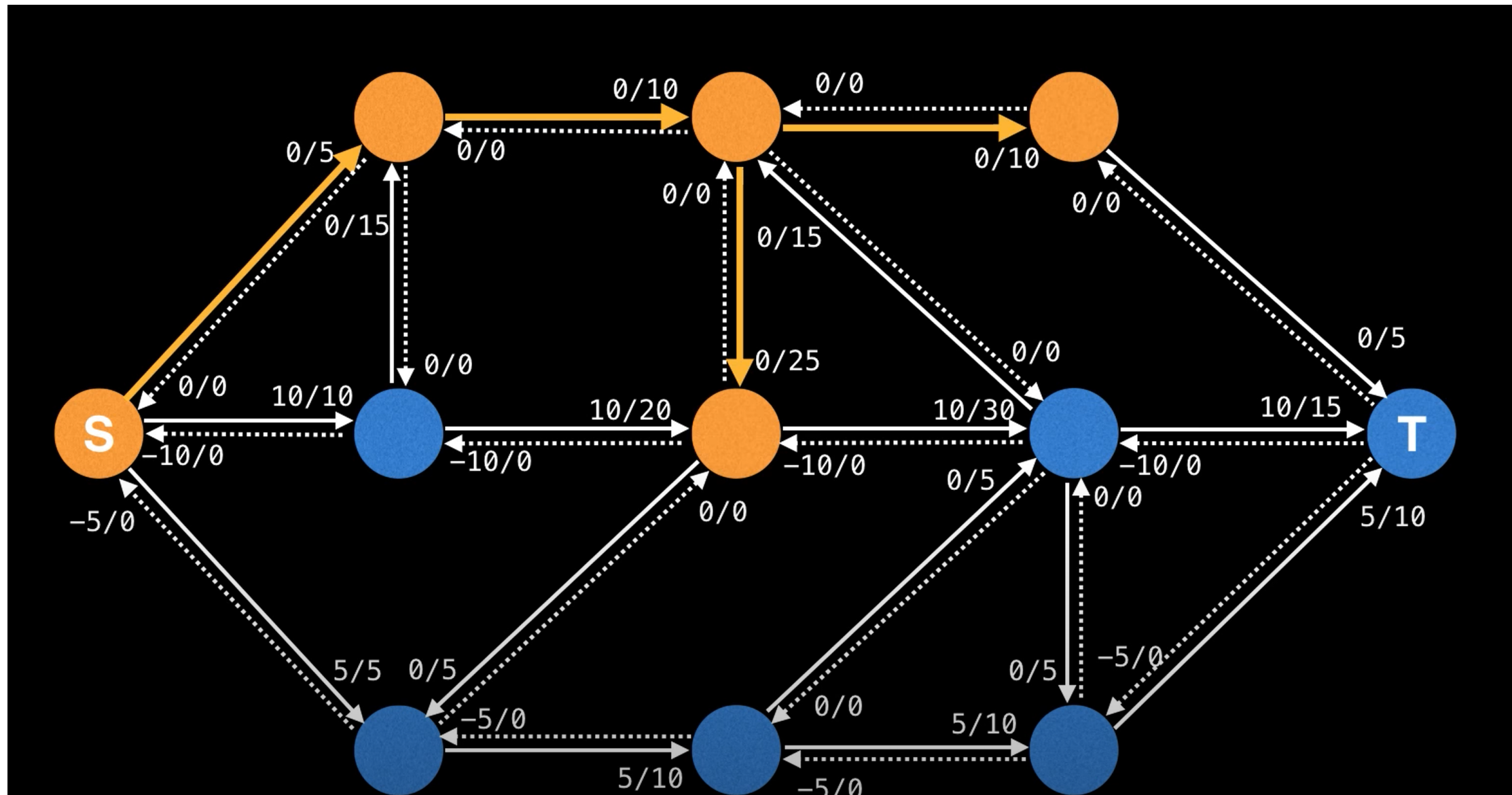


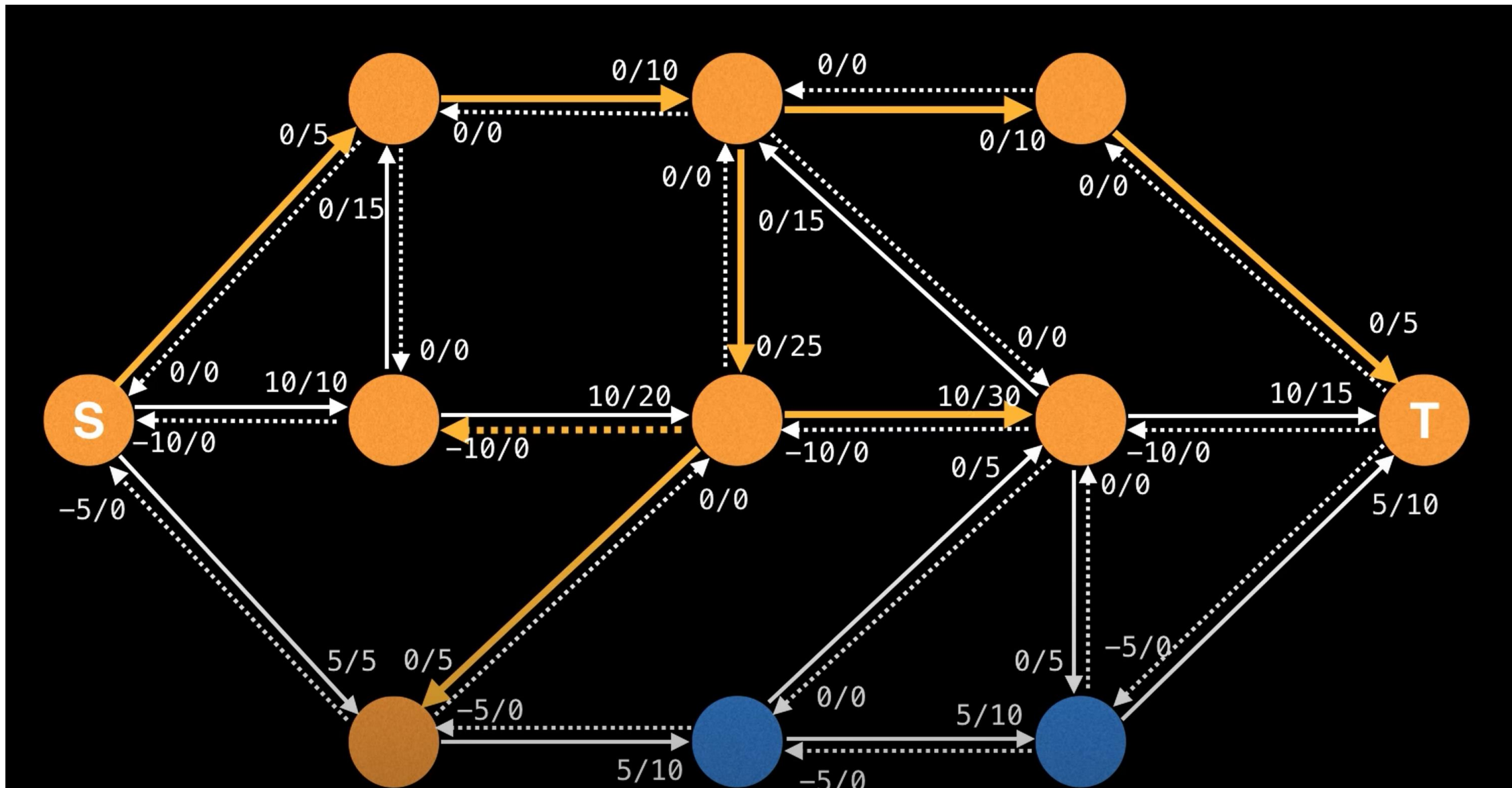


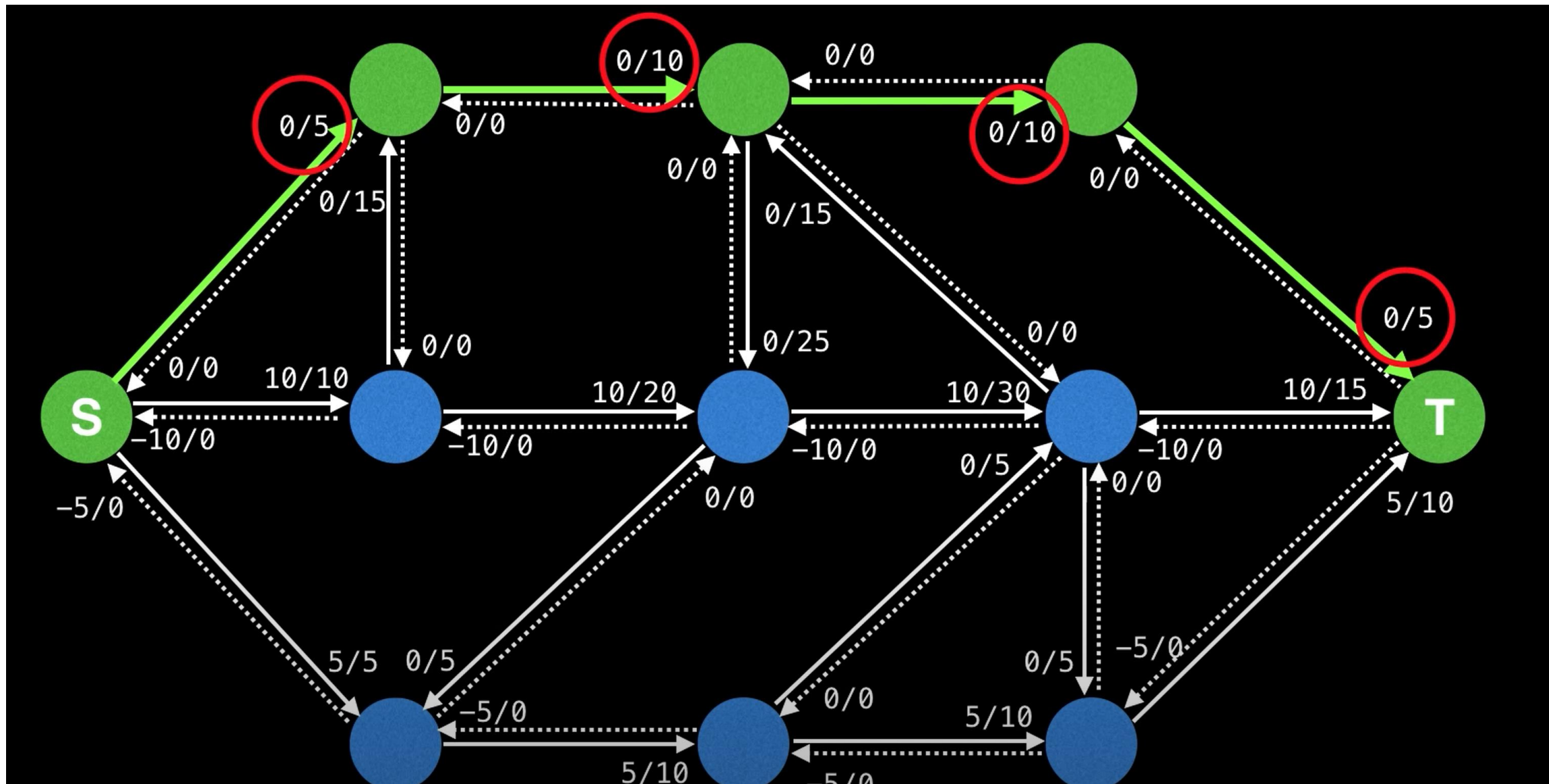












Max flow = 5 + 10 + 5 = 20

